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point P_2 where its position vector is $\vec{r}_2 = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$.

The displacement vector is given by

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$
$$= (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}.$$

This displacement is also shown in Figure 2.27.

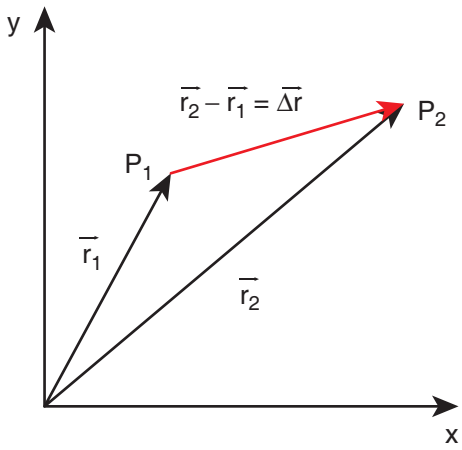
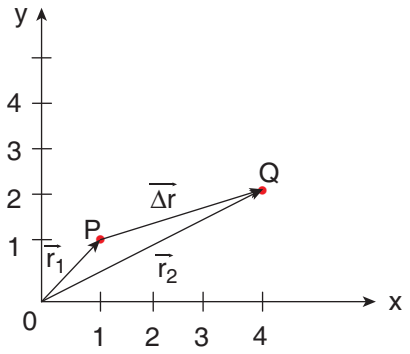


Figure 2.27 Displacement vector

EXAMPLE 2.17

Calculate the displacement vector for a particle moving from a point P to Q as shown below. Calculate the magnitude of displacement.



Solution

The displacement vector $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$, with

$$\vec{r}_1 = \hat{i} + \hat{j} \text{ and } \vec{r}_2 = 4\hat{i} + 2\hat{j}$$
$$\therefore \Delta\vec{r} = \vec{r}_2 - \vec{r}_1 = \left(4\hat{i} + 2\hat{j}\right) - \left(\hat{i} + \hat{j}\right)$$
$$= (4-1)\hat{i} + (2-1)\hat{j}$$
$$\therefore \Delta\vec{r} = 3\hat{i} + \hat{j}$$

The magnitude of the displacement vector $\Delta r = \sqrt{3^2 + 1^2} = \sqrt{10}$ unit.

Note

(1) The Distance travelled by an object in motion in a given time is never negative or zero, it is always positive.

(2) The displacement of an object, in a given time can be positive, zero or negative.

(3) The displacement of an object can be equal or less than the distance travelled but never greater than distance travelled.

(4) The distance covered by an object between two positions can have many values, but the displacement between them has only one value (in magnitude).

2.8 DIFFERENTIAL CALCULUS

The Concept of a function

1) Any physical quantity is represented by a "function" in mathematics. Take the example of temperature T. We know that the temperature of the surroundings is changing throughout the day. It increases till noon and decreases in the evening. At any